SMART WEARABLE & HUD OPTICS | PROJECT OVERVIEW

Industrial Sound Intelligence for Smart Wearables

AI-powered machine health monitoring and hands-free industrial voice assistance on a unified edge wearable platform.

CORE ARCHITECTURAL PROPOSITION: A single wearable dual-microphone array captures acoustic data for two parallel AI engines: passive machine condition listening and active hands-free worker dialogue.

PILLAR 1: ACOUSTIC MACHINE MONITORING

1. Audio Acquisition & Validation
Continuous 16 kHz acoustic sensing capturing machinery signatures (motors, pumps, bearings, compressors).

2. 180-Dimensional Feature Extraction
Time-domain (RMS, Crest Factor, Kurtosis), spectral descriptors (Wiener Flatness, HFER), and 128 Log-Mel & 40 MFCC stats.

3. Eight-Class Sound Event Detection (SED)
Classifies sound into 8 categories (Normal, Alarms, Leaks, Cavitation) with Shannon entropy uncertainty thresholding.

4. Asset-Specific Anomaly Detection
Evaluates Mahalanobis distance against calibrated asset baselines with Ledoit-Wolf diagonal covariance shrinkage.

5. Physical Defect Attribution
Provides diagnostic evidence for bearing spalling, cavitation micro-implosions, mechanical unbalance, and friction.

PILLAR 2: INDUSTRIAL VOICE ASSISTANT

1. Voice Capture & Quality Diagnostics
Wearable microphone input with real-time SNR estimation, clipping detection, and dynamic noise profiling.

2. Streaming Voice Activity Detection (VAD)
Silero VAD 32ms frame tracking preserving plosive consonants with configurable speech boundary hysteresis.

3. Faster-Whisper ASR (Quantized INT8)
High-speed CPU speech-to-text (~185 ms) conditioned with industrial vocabulary biasing and plant equipment hotwords.

4. 12 Industrial Intents & Entity Extraction
Extracts operational intents (SOP, torque, temperature) and slots (machine IDs, components, units) with 100% accuracy.

5. Grounded OEM RAG & Technical TTS
Answers retrieved from 20 approved OEM manuals (106 chunks) and spoken via phonetic unit-normalizing speech synthesis.

⚠ **CRITICAL SAFETY CROSS-LINK:** Speech Interference Gate (`is_suppressed_by_speech = True`) automatically suspends machinery anomaly scoring while the operator speaks, mathematically preventing conversational voice from triggering false machine alarms.

KEY TAKEAWAY One wearable audio interface supports both continuous machine-condition awareness and hands-free maintenance assistance.

TECHNICAL ARCHITECTURE | DEEP IMPLEMENTATION DETAILS

From Raw Audio to Actionable Intelligence

Two parallel mathematical signal processing and neural inference pipelines running on edge hardware.

ACOUSTIC MACHINE MONITORING PIPELINE

1. Audio Ingestion & Preprocessing

- 16 kHz float32 normalization with payload boundary limits (≤ 50MB, ≤ 180s).
- NaN/Inf sanitization with clipping detection (> 99% peak) & low-RMS silence rejection.

2. 180-Dimensional Feature Extraction

- **Time-Domain (5):** RMS energy, Crest Factor, Kurtosis, Skewness, Zero-Crossing Rate.
- **Spectral (7):** Centroid, Spread, Wiener Flatness, Roll-off (85%), Flux, HFER (>4 kHz), Dominant Peak.
- **Time-Frequency (168):** 64 Log-Mel mean+std (128) & 20 MFCCs mean+std (40).

3. 8-Class Sound Event Detection (SED)

- Classes: `MACHINERY_NORMAL, SPEECH, ALARM, IMPACT, LEAK, CAVITATION, SILENCE, UNKNOWN`.
- Shannon Entropy Gate: Ambiguous signals (H > 1.70 nats) routed to `UNKNOWN`.
- **Verified Benchmark:** Macro-F1 = 97.5% (Precision: 97.9%, Recall: 97.5% on N=160 held-out samples).

4. Asset Anomaly Mahalanobis Engine

- Ledoit-Wolf diagonal covariance shrinkage (λ = 0.05) guaranteeing non-singular precision.
- Continuous score s in [0, 1] with 4-frame rolling median persistence filter.
- **Verified Benchmark:** ROC-AUC = 0.946–0.975 | pAUC(0.1) = 0.929 | 0.00 False Alarms / hr.

5. Diagnostic Physical Fault Attribution

- Structured attribution for bearing spalling, cavitation micro-implosions, unbalance, and friction.
- **Latency:** Continuous acoustic pipeline executes in **1.41 ms total** (Extraction: 1.32ms, SED: 0.03ms, Anom: 0.06ms).

INDUSTRIAL SPEECH INTELLIGENCE PIPELINE

1. Audio Quality Diagnostics & VAD

- Real-time SNR estimation, peak clipping detection, and dynamic background noise profiling.
- Streaming Silero VAD (32ms / 512-sample frame tracking) preserving plosive consonants.

2. Faster-Whisper ASR (Quantized INT8)

- Hardware-aware INT8 CPU execution (~185 ms latency) with automatic CUDA fallback.
- Contextual initial prompt biasing & industrial hotwords (MTR-03, PMP-01A, SKF 6308, LOTO).
- **Robustness:** 2.4% WER in clean audio to 16.2% WER under harsh 0 dB plant noise.

3. 12 Industrial Intents & Entity Extraction

- 12 Intents: `GET_SOP, CHECK_TEMPERATURE, CHECK_ACOUSTIC_ANOMALY, GET_TORQUE_SPEC, CHECK_ASSET_STATUS, IDENTIFY_COMPONENT, REPORT_FAULT, EXPLAIN_ALARM, NEXT_STEP, REPEAT, CANCEL, EMERGENCY_HELP`.
- **Accuracy:** 100.0% verified classification accuracy on canonical test suite.
- Slot extraction: Machine tags (MTR-03), components, torque (Nm), temperature (°C).

4. Safety Router & Grounded OEM RAG

- Centralized Safety Router: Emergency overrides bypass retrieval; supervisor role gates state changes.
- RAG Corpus: Ingests 20 approved OEM technical manuals & SOPs (106 chunks indexed).
- Phonetic TTS expands technical terms: *MTR-03 → "M-T-R zero three"*, *45 Nm → "45 Newton meters"*.

> **Execution Walkthrough:**
> Operator: *"Check MTR-03 casing temperature."*
> Faster-Whisper STT → Intent: `CHECK_TEMPERATURE` → Entity: `asset_id: MTR-03` → Verified Telemetry → Spoken Phonetic Response: *"M-T-R zero three casing temperature is eighty-three point six degrees Celsius."*

KEY TAKEAWAY **Acoustic monitoring detects unusual machine sounds, while speech intelligence enables hands-free interaction with industrial information.**

OPERATIONAL IMPACT | FIELD DEPLOYMENT & VALUE

From Sound Detection to Safer Industrial Maintenance

End-to-end incident walkthrough, safety architecture boundaries, and measured engineering validation.

OPERATIONAL INCIDENT WALKTHROUGH: ABNORMAL PUMP CAVITATION DETECTION & ASSISTANCE

1. LISTEN

Wearable mic passively captures pump PMP-01A acoustic emissions (16kHz).

2. ANALYZE

Feature extractor spots high-frequency energy ratio rise (HFER ≥ 0.22).

3. COMPARE

Mahalanobis distance vs PMP-01A baseline (s = 0.91 > 0.88 critical alarm).

4. ATTRIBUTE

System flags CAVITATION_OR_TURBULENCE; alerts operator via AR HUD optics.

5. ASSIST

Technician speaks: 'Show pump inspection SOP.' Voice intent GET_SOP resolved.

6. RESPOND

RAG retrieves approved manual DOC-PMP-SOP-002, speaking Step 1 LOTO isolation.

SAFETY & RELIABILITY GOVERNANCE

- **Diagnostic Separation:** Acoustic patterns are probabilistic indicators, never physical certainty. Human inspection is required.
- **Hardwired E-Stop Independence:** AI assistant is strictly advisory. Certified physical emergency-stop circuits remain independent safety relays.
- **Zero-Hallucination Grounding:** Procedural answers are cited from approved OEM manuals; unverified torque values are rejected.
- **Operator Confirmation Gates:** State-changing operations (fault reporting, next-step procedure) require explicit worker confirmation.

MEASURED ENGINEERING BENCHMARKS

99 / 99 Tests Passing (100%)
Zero regressions across acoustic, speech, thermal, and RAG suites.

1.41 ms Continuous Pipeline Latency
Extraction (1.32ms), SED (0.03ms), Mahalanobis Anomaly (0.06ms).

97.5% SED Macro-F1 | 0.975 Anomaly AUC
N=160 held-out samples, 0.00 false alarms/operating hr with persistence.

100.0% Intent Accuracy | 20 OEM Manuals
12 canonical intents validated; 106 grounded technical chunks indexed.

PRACTICAL INDUSTRIAL VALUE

- **Hands-Free Field Mobility:** Eliminates hazardous manual tablet handling in rotating machinery and high-voltage zones.
- **Early Incipient Fault Warning:** Detects pump cavitation and bearing race spalling weeks before thermal overheating occurs.
- **Alert Fatigue Elimination:** Speech interference and transient tool clatter are filtered out to preserve operator trust.
- **Complete Audit Compliance:** All safety directives, threshold alerts, and operator queries logged to immutable audit trails.

KEY TAKEAWAY **The goal is to combine machine sound awareness with hands-free AI assistance to support industrial workers during inspection and maintenance.**